



RAMAIAH

Institute of Technology

Department of Computer Science and Engineering

**Fundamentals of Computing & C
Programming Laboratory
Subject Code: CSL18/CSL28
Subject credits 0:1:1
I/II Semester
January - May 2020**



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Algorithms and Flowchart

Algorithm:

Definition: An algorithm is a sequence of steps to solve a particular problem. Algorithm is first step of the solution process, after the analysis of problem.

Example: Write an algorithm to find out number is odd or even.

Answer

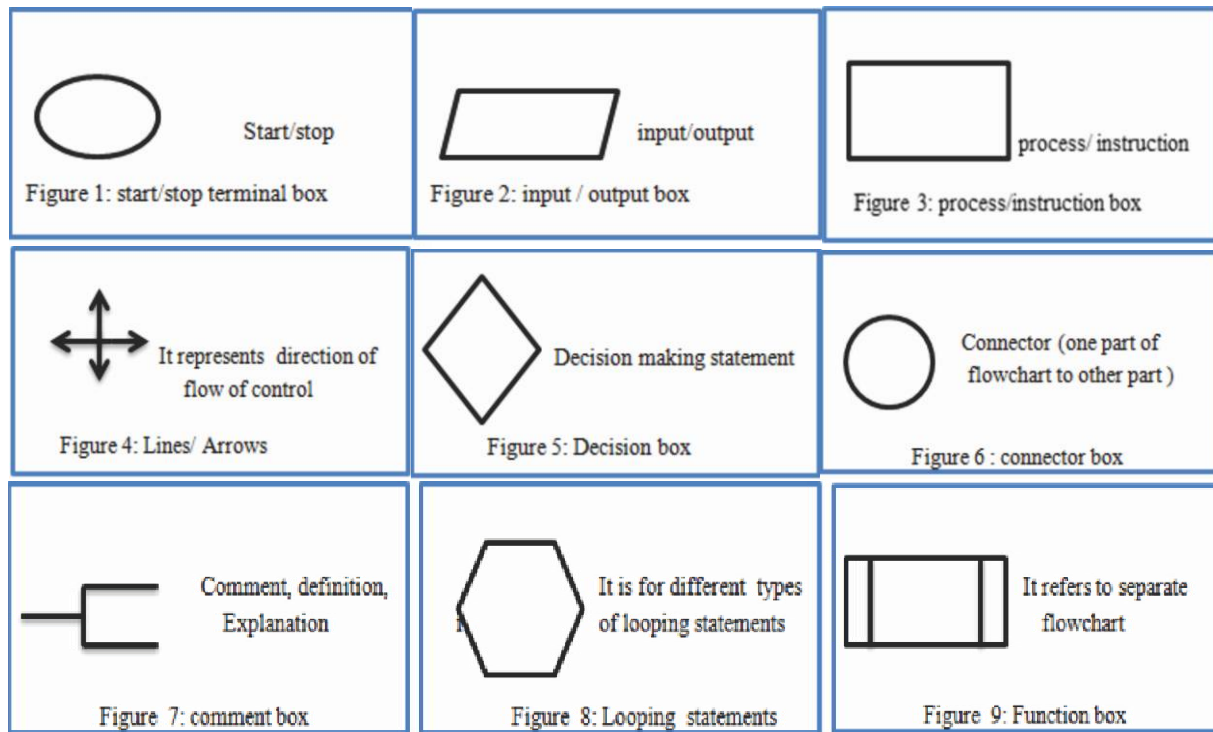
```
Step 1: start
Step 2: input number
Step 3: rem=number mod 2
Step 4: if rem=0 then
    print "number is even"
else
    print "number is odd"
end if
Step 5: stop
```

Flowchart:

Definition: Graphical representation of any program is called flowchart.

Example: Draw a flowchart to find out number is odd or even

There are some standard graphics that are used in flowchart as following



Basic Concepts of a C Program
Structure of C Program

Documentation Section
Link Section
Definition Section
Global Declaration Section
main() function section { Declaration part Executable part }
Subroutine Section Function 1 Function 2 : : Function n

C Identifiers

Identifier refers to name given to entities such as variables, functions, structures etc. Identifier must be unique.

For example: `int money;`

`double Account_balance;`

Here, `money` and `Account_balance` are identifiers.

C Data Types

Type	Contains	Default	Size	Range
boolean	true or false	false	1 bit	NA
char	Unicode character	\u0000	16 bits	\u0000 to \uFFFF
byte	Signed integer	0	8 bits	-128 to 127
short	Signed integer	0	16 bits	-32768 to 32767
int	Signed integer	0	32 bits	-2147483648 to 2147483647
long	Signed integer	0	64 bits	-9223372036854775808 to 9223372036854775807
float	IEEE 754 floating point	0.0	32 bits	$\pm 1.4\text{E-}45$ to $\pm 3.4028235\text{E}+38$
double	IEEE 754 floating point	0.0	64 bits	$\pm 4.9\text{E-}324$ to $\pm 1.7976931348623157\text{E}+308$

Input and Output Statements

1) scanf()

Syntax: `scanf("format specifiers", &value1, &value2.....);`

Example: `int a;`
`float b;`
`scanf("%d%f", &a, &b);`

2) printf()

Syntax: `printf("User defined Message");`
`printf("Format specifiers", value1, value2);`

Example: `int a=10;`
`double d=13.4;`
`printf("%f%d", d, a);`

Format specifiers in C:

Format specifier	Type of value		
%d	integer	%u	unsigned int
%f	float	%ld	long int
%lf	double	%lf	long double
%c	Single character		
%s	String		

Tutorial 1
Computing System and C data types, variables, tokens and symbolic constants

I. Indicate whether the following statements are true or false.

1. A CPU consists of ALU, Memory and Control Unit.
2. Linking and loading is the step that prepares the object program for execution.
3. An algorithm describes the problem solution step by step, while a computer program solves the problem in one step.
4. A computer program is the implementation of an algorithm.
5. A microprocessor is a process that is very small.
6. Spreadsheets are useful to manipulate objects graphically.
7. Computer aided design is used only to design microcomputers.
8. The fastest computer is microprocessor.
9. Spaces cannot be present within a variable name.
10. Variables can be declared anywhere in a C program.

II. Choose the best option to complete each statement:

1. Instructions and data are stored in
 - a. The arithmetic logic unit (ALU)
 - b. The control unit (processor)
 - c. The central processing unit (CPU)
 - d. The memory
 - e. The keyboard
2. Source code is
 - a. The result of compiler operations
 - b. The process of getting information from the processor.
 - c. The set of instructions in a computer language that solves a specific problem.
 - d. The data stored in computer memory.
3. Object code is
 - a. The result of compiler operations on the source code.
 - b. The process of obtaining information from the processor.
 - c. A computer program.
 - d. The result of the linking/loading process.
4. Which of the following is not a valid variable name declaration?
 - a) int _a3;
 - b) int a_3;
 - c) int 3_a;
 - d) int _3a

5. Which of following is not a valid name for a C variable?
- Examveda
 - Exam_veda
 - Exam veda
 - None of these

III. Fill in the blanks:

- _____ is set of instructions that tells a computer what to do
- _____ is the brain of a computer
- An _____ describes the problem solution step by step
- _____ is a software tool designed to work with data stored in a grid or a table
- _____ acts as an interface between the user and the hardware.
- _____ are special words whose meaning is known to the compiler.
- # define is a _____
- Program Execution in C begins at _____
- Every program statement in C must end with a _____
- The smallest individual units in C are called _____
- C programs are converted into machine language with the help of _____
- _____ is the only function all C programs must contain.
- The processor and ALU together are called _____
- _____ Language instructions are written as sequence of 0's and 1's.
- _____ Language instructions are written in symbolic statements instead of binary.
- _____ are computer languages that have English like commands.
- _____ is the set of grammatical rules used to govern the writing of software in a programming language.
- The process of removing errors/bugs from the program is called _____
- The process of converting an assembly language program to binary is performed by an _____
- The phases of software life cycle are _____, _____, _____, _____, _____

IV. Which of the following are invalid variable names and why?

- First_tag
- char
- average_number
- group one
- int_type

6. `_total`
7. `%name`
8. `$name`
9. `3rd_cross`
10. `New year`

V. Find errors in the following #define (symbolic constant) statements.

1. `#define X =25`
2. `# define MAX 50`
3. `#define N 25;`
4. `#define A 10, B 20`
5. `#Define MIN 10`

Tutorial 2

Operators and Evaluation of expressions

C Operator Precedence Table

C operators are listed in order of *precedence* (highest to lowest). Their *associativity* indicates in what order operators of equal precedence in an expression are applied.

Operator	Description	Associativity	Rank
() []	Function call Array element reference	Left to Right	1
+ - ++ -- ! ~ * & sizeof (type)	Unary Plus Unary Minus Increment Decrement Logical negation Ones complement Pointer reference (indirection) Address Size of an object Type cast	Right to Left	2
* / %	Multiplication Division Modulus	Left to Right	3
+ -	Addition, Subtraction	Left to Right	4
<< >>	Left Shift Right Shift	Left to Right	5
< <= > >=	Less than, Less than or equal to Greater than, Greater than or equal to	Left to Right	6
= = !=	Equality Inequality	Left to Right	7
&	Bitwise AND	Left to Right	8
^	Bitwise XOR	Left to Right	9
	Bitwise OR	Left to Right	10
&&	Logical AND	Left to Right	11
	Logical OR	Left to Right	12
?:	Conditional Expression	Right to Left	13
*= /= %= += -= &= ^= = = <<= >>=	Assignment Operators	Right to Left	14
,	Comma Operator	Left to Right	15

Note 1:

Parentheses are also used to group sub-expressions to force a different precedence; such parenthetical expressions can be nested and are evaluated from inner to outer.

Note 2:

Postfix increment/decrement have high precedence, but the actual increment or decrement of the operand is delayed (to be accomplished sometime before the statement completes execution). So in the statement $y = x * z++$; the current value of z is used to evaluate the expression (i.e., $z++$ evaluates to z) and z only incremented after all other statements are executed.

1. Evaluate the following expressions.

a. $X = 3 * 4 + 5 * 6$	b. $X = 3 * (4 + 5) * 6$
c. $X = 3 * 4 \% 5 / 2$	d. $X = 3 * (4 \% 5) / 2$
e. $X = 3 * ((4 \% 5) / 2)$	f. $X = 17 - 8 / 4 * 2 + 3 - ++a$ (assume $a=6$)

2. Assume the following variables declaration,

int a = 0, b = 1, c = -1;

float x = 2.5, y = 0.0;

Evaluate the following expressions:

a. $a += b - c * 10$	b. $--a * (5 + b) / 2 - c ++ * b$
----------------------	-----------------------------------

c. $a * b * c$	d. $(x > y) + !a \parallel c ++$
e. $a < b \&\& c < b$	f. $b + c \parallel !a$
g. $x * 5 \&\& 5 \parallel (b / c)$	h. $a <= 10 \&\& x >= 1 \&\& b$
i. $!x \parallel !c \parallel b + c$	j. $x * y < a + b \parallel c$

3. Find the output for the following code segments:

a.

```
int a=27,b=6,c;  
c=b%a;  
printf("The value of c=%d",c);
```

Answer:

- b. `int a=27,b=6;`
`float c;`
`c=a/(float)b;`
`printf("The value of c=%f",c);` **Answer:**
- c. `int a;`
`float b=6,c=18.6;`
`a=c/b;`
`printf("the value of a=%d",a);` **Answer:**
- d. `int b=6;`
`float a, c=18.6;`
`a= (int) c/b;`
`printf("the value of a=%f",a);` **Answer:**

4. Convert the following equations into corresponding C statements.

1.
$$Z = \frac{8.8(a+b)^2/c - 0.5 + 2a/(q+r)}{(a+b)*(1/m)}$$

2.
$$X = \frac{-b + (b*b) + 4ac}{2a}$$

3.
$$R = \frac{2v + 6.22(c+d)}{g+v}$$

4.
$$A = \frac{7.7b(xy+a)/c - 0.8 + 2b}{(x+a)(1/y)}$$

5. Give a memory snapshot after each statement is executed, assuming that x is equal to 2 and that y is equal to 4 before the statement is executed. Also, assume that all the variable are integers:

$z = x++ * y;$	x	y	z
Initial Values			
After Execution			

$z = ++x * y;$	x	y	z
Initial Values			
After Execution			

$x += y;$	x	y
Initial Values		
After Execution		

$y \% = x;$	x	y
Initial Values		
After Execution		

6. Which of the following expressions are true?

a. $!(5 + 5 >= 10)$	b. $5 + 5 == 10 \parallel 1 + 3 == 5$
c. $5 > 10 \parallel 10 < 20 \&\& 3 < 5$	d. $10 != 15 \&\& !(10 < 20) \parallel 15 > 30$

Tutorial 3**Evaluation of expressions contd., Mathematical Functions, Type Conversions****1. Determine the result of each of the following logical expressions****if a=5, b=10 and c= - 6.**

a. <code>a > b && a < c</code>	b. <code>a < b && a > c</code>
c. <code>a == c b > a</code>	d. <code>b > 15 && c < 0 a > 0</code>

2. Compute the output of the following programs:

1. <code>#include<stdio.h></code> <code>int main()</code> <code>{</code> <code>int x=3,y=5,z=7;</code> <code>int a,b;</code> <code>a=x*2+y/5-z*y;</code> <code>b=++x*(y-3)/2-z++*y;</code> <code>printf("\n a=%d",a);</code> <code>printf("\n b=%d",b);</code> <code>return 0;</code> <code>}</code> Answer:	2. <code>#include<stdio.h></code> <code>int main()</code> <code>{</code> <code>int num=7;</code> <code>printf("%d\n",num++);</code> <code>num=num+1;</code> <code>printf("%d\n",++num);</code> <code>return 0;</code> <code>}</code> Answer:
---	--

```
3. #include<stdio.h>
int main( )
{
    int a=4,b=5,c=6;
    a=b==c;
    printf("\n a=%d", a);
    return 0;
}
```

Answer:

```
4. #include<stdio.h>
int main( )
{
    int a=100,b=3;
    float c;
    c=a/b;
    printf("\n c=%f", c);
    return 0;
}
```

Answer:

```
5. #include<stdio.h>
int main( )
{
    int a=2,b=3,c,d;
    c=a++;
    d=++b;
    printf("\n c=%d d=%d",c,d);
    return 0;
}
```

Answer:

```
6. #include<stdio.h>
int main( )
{
    int a=2,b=3;
    printf("\n %d",++a - b);
    return 0;
}
```

Answer:**3. Find the output for the following code snippets:**

```
1) #include<stdio.h>
int main( )
{
    int sum=17,count=5;
    double mean;
    mean=(double)sum/count;
    printf("Value of mean:%f\n",mean);
    return 0;
}
```

Answer:

```
2) #include<stdio.h>
int main( )
{
    int i=17;
    char c='c';
    int sum;
    sum=i+c;
    printf("Value of sum:%d\n",sum);
    return 0;
}
```

Answer:

```
3) #include<stdio.h>
int main( )
{
    int i=17;
    char c='c';
    float sum;
    sum=i+c;
    printf("The value of sum:%f\n",sum);
    return 0;

}
```

Answer:

```
4) #include<stdio.h>
int main( )
{
    int a=100,b=3;
    float c;
    c=a/b;
    printf("\n c=%f",c);
    return 0;
}
```

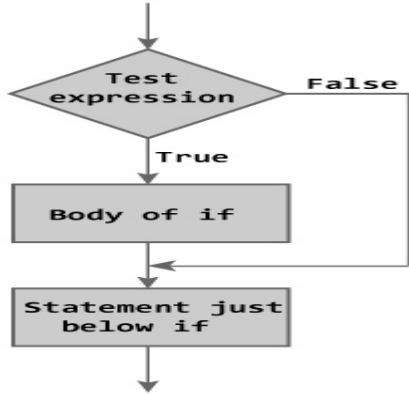
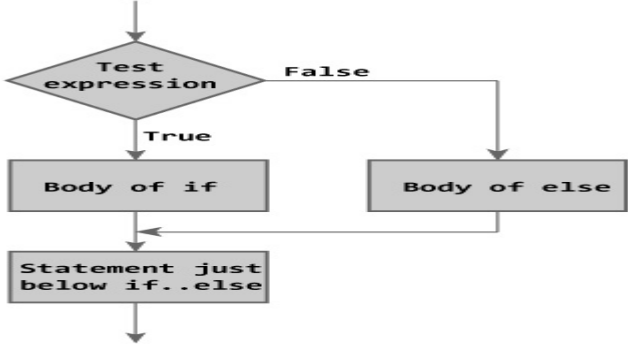
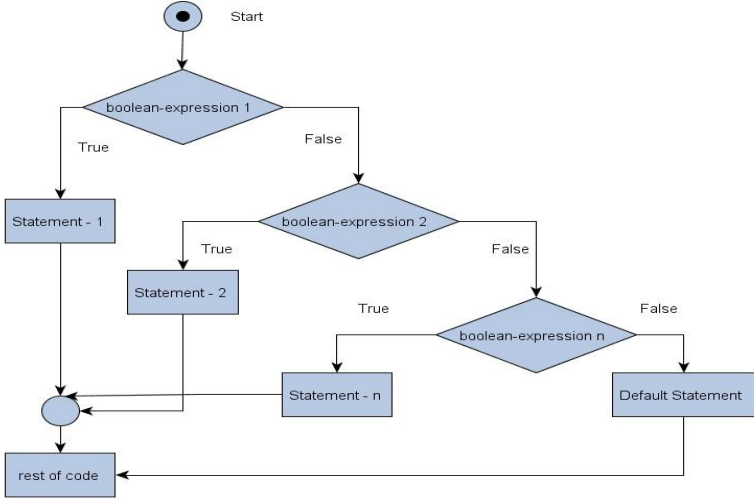
Answer:

4. Evaluate the following expressions:

- | | |
|----------------------|----------------|
| a. floor(-2.6) | Answer: |
| b. pow(2,-3) | Answer: |
| c. sqrt(floor(10.7)) | Answer: |
| d. fabs(-10*2.5) | Answer: |
| e. fabs(pow(-2,5)) | Answer: |

Tutorial 4 if statement, if else statement, nested else if statements

Syntax and Flow Chart of if, if-else and else-if ladder statements

Syntax	Flowchart
Simple if <pre> if (testExpression) { // statements } </pre>	 Figure: Flowchart of if Statement
if else <pre> if (testExpression) { // codes inside the body of if } else { // codes inside the body of else } </pre>	 Figure: Flowchart of if...else Statement
else if ladder <pre> if (Test Condition -1) { Statement -1; } else if (Test Condition -2) { Statement -2; } else if (Test Condition -3) { Statement -3; } </pre>	 Else-if Ladder statement flow chart

1. Find errors, if any, in each of the following segments:

a. <code>if(x + y = z && y > 0)</code> <code>printf(" ");</code>
b. <code>if(code > 1);</code> <code>a=b+c</code> <code>else</code> <code>a=0;</code>
c. <code>if(p<0) (q<0)</code> <code>printf("sign is negative");</code>

2. Assuming that x=5, y=0, and z=1 initially, what will be their values after executing the following code segments?

a) <code>if (x && y)</code> <code>x=10;</code> <code>else</code> <code>y =10;</code> Answer:	b) <code>if (x y z)</code> <code>y=10;</code> <code>else</code> <code>z =0;</code> Answer:
c) <code>if (x)</code> <code>if (y)</code> <code>z = 10;</code> <code>else</code> <code>z = 0;</code> Answer:	d) <code>if (x == 0 x && y)</code> <code>if(!y)</code> <code>z = 0;</code> <code>else</code> <code>y = 1;</code> Answer:

3. Find the output for the following programs:

<pre>a. #include<stdio.h> int main() { int a=2,b=3,c=4; if(b==2) a=10; else c=10; printf("\n a=%d \t b=%d \t c=%d",a,b,c); return 0; }</pre> Answer:	<pre>b. #include<stdio.h> int main() { int a=2,b=3,c=4; if(a&& b) c=10; else c=20; printf("\n a=%d \t b=%d \t c=%d",a,b,c); return 0; }</pre> Answer:
<pre>c. #include<stdio.h> int main() { int a=2,b=3,c=4; if(a==0 b>=c && c> 0) if(a && b) c=10; else c=20; printf("\n a=%d \t b=%d \t c=%d",a,b,c); return 0; }</pre> Answer:	<pre>d. #include<stdio.h> int main() { int a=2,b=3,c=4; if(a=b<c) { c++; a--; } ++b; printf("\n a=%d \t b=%d \t c=%d",a,b,c); return 0; }</pre> Answer:
<pre>e. #include<stdio.h> int main() { int a=3; if(a<10) printf("\n Less"); if(a<20) printf("\n Less"); if(a<30) printf("\n Less"); return 0; }</pre> Answer:	<pre>f. #include<stdio.h> int main () { int a=10,b=20,c=30,d=40; if(c<d) if(c<b) printf("\n c"); else if(a<c) printf("\n a"); if(a>b) printf("\n b"); else printf("\n d"); } return 0;</pre> Answer:

```
g. #include<stdio.h>
int main( )
{
    printf("HELLO");
    if(-1)
        printf("WORLD");
    return 0;
}
```

Answer:

```
h. #include <stdio.h>
int main()
{
    int x = 0;
    if (x == 0)
        printf("hi");
    else
        printf("how are u");
    printf("hello");
    return 0;
}
```

Answer:

```
i. #include<stdio.h>
int main( )
{
    int k, num = 100;
    k = (num > 50 ? (num <= 10 ? 100 : 200):
500);
    printf("%d\t%d\n", k, num);
    return 0;
}
```

Answer:

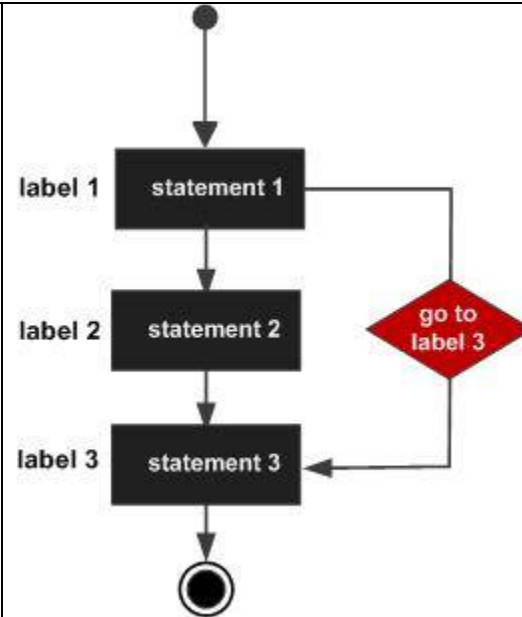
Tutorial 5
switch case statement, goto statement

Syntax and Flowchart of Switch and goto statements

Syntax	Flowchart
Switch statement <pre> switch (expression) { case value1 : statement-list1; break; case value2 : statement-list2; break; case value3 : statement-list3; break; case ... } </pre>	

Goto statement

```
goto label;
..
.
label: statement;
```

**1. Find the output of the following code snippet.**

```
#include<stdio.h>
main()
{
    int i=1;
    Back: printf("INDIA\n");
    if(i!=5)
    {
        i++;
        goto Back;
    }
}
```

Output:**2. Each case statement in switch is separated by**

a) break b) continue c) exit d) goto

3. Write the output for below C code.

```
#include <stdio.h>
int main( )
{
    char suite = 3 ;
    switch ( suite )
    {
        case 1 :
            printf ( "\nDiamond" ) ;
```

```
    case 2 :  
        printf ( "\nSpade" ) ;  
    default :  
        printf ( "\nHeart" ) ;  
    } printf ( "\n Club" ) ;  
    return 0; }
```

Output:

4. Write the output for below C code.

```
#include <stdio.h>  
#define a 1  
#define b 2  
int main( )  
{  
    int x = 1;  
    switch (x)  
    {  
        case a:  
            printf("yes ");  
        case b:  
            printf("no\n");  
            break;  
    }  
    return 0;  
}
```

a) yes no b) yes c) no d) Compile time error

5. Write the output for below C code.

```
#include <stdio.h>  
int main( )  
{  
    int k, j = 2 ;  
    switch ( k = j + 1 )  
    {  
        case 0 :  
            printf ( "\nTailor" ) ;  
        case 1 :  
            printf ( "\nTutor" ) ;  
        case 2 :  
            printf ( "\nTramp" ) ;  
        default :  
            printf ( "\nEngineer" ) ;  
    }  
    return 0; }
```

Output:

6. Write the output for below C code.

```
#include <stdio.h>
int main( )
{
    int k ;
    float j = 2.0 ;
    switch ( k = j + 1 )
    {
        case 3 :
            printf ( "\nTrapped" ) ;
            break ;
        default :
            printf ( "\nCaught!" ) ;
    } return 0;
}
```

Output:**7. Write the output for below C code.**

```
#include <stdio.h>
int main( )
{
    int ch = 'a' + 'b' ;
    switch ( ch )
    {
        case 'a' :
        case 'b' :
            printf ( "\nYou entered b" ) ;
        case 'A' :
            printf ( "\na as in ashar" ) ;
        case 'b' + 'a' :
            printf ( "\nYou entered a and b" ) ;
    }
    return 0;
}
```

Output:**8. Write the output for below C code.**

```
#include <stdio.h>
int main()
{
```

```
int temp = 100;
switch(temp)
{
    case 10 : printf("10");
              break;
    case 10 : printf("10"); // won't compile!
              break;
    case 100 : printf("1000");
              break;
    default : printf("default");
             break;
}
return 0;
}
```

Output:

9. Which of the following are valid usages of case labels.

- i. case 1+1:
- ii. case 'A':
- iii. case 67:
- iv. case var:
- v. case num1:
- vi. case 10:
- vii. case 20+20:
- viii. case 10.12:
- ix. case 7.5:

10. Write the output for below C code.

```
#include<stdio.h>
int main( )
{
    int check=2;
    switch(check)
    {
        case 1: printf("Hi");
        case 2: printf(" Hello");
        case 3: printf(" How r u?");
        default: printf(" All d best");
    }
    return 0;
}
```

Output:

Tutorial 6 for loop, while loop, do-while loop

Syntax and Flowchart of for loop, while loop, do while loop

Syntax	Flowchart
While loop: <pre>while(condition) { statement(s); }</pre>	<pre> graph TD Start(()) --> WhileCode[while(condition) { conditional code ; }] WhileCode --> Condition{condition} Condition -- "If condition is true" --> CodeBlock[code block] CodeBlock --> Condition Condition -- "If condition is false" --> End((())) </pre>
For loop: <pre>for (init; condition; increment) { statement(s); }</pre>	<pre> graph TD Start(()) --> ForCode[for(init; condition; increment) { conditional code ; }] ForCode --> Init[Init] Init --> Condition{condition} Condition -- "If condition is true" --> CodeBlock[code block] CodeBlock --> Increment[Increment] Increment --> Condition Condition -- "If condition is false" --> End((())) </pre>
do-while loop: <pre>do { statement(s); } while(condition);</pre>	<pre> graph TD Start(()) --> DoWhileCode[do { conditional code ; } while (condition)] DoWhileCode --> CodeBlock[code block] CodeBlock --> Condition{condition} Condition -- "If condition is true" --> CodeBlock Condition -- "If condition is false" --> End((())) </pre>

1. Find the output of the following program code.

<pre> 1) #include <stdio.h> int main() { int val=1; do{ val++; ++val; }while(val++>25); printf(“%d\n”,val); return 0; } </pre> Answer:	<pre> 2) #include<stdio.h> int main() { int num=10; for(--num;) printf(“%d”,num); return 0; } </pre> Answer:
<pre> 3) int main() { int i; for(i=0;i<10;i++); printf(“%d “,i); return 0; } </pre> Answer:	<pre> 4) int main() { int i = 0,j; for (j = 1; j < 7; j++) { i = i + 1; } printf(“%d”, i); return 0; } </pre> Answer:
<pre> 5) #include<stdio.h> int main() { int i=0; do { printf(“%d”,i); i++; }while(i<=10); printf(“\n stop”); return 0; } </pre> Answer:	<pre> 6) #include<stdio.h> int main() { int num=10; for(;;) printf(“Hi”); return 0; } </pre> Answer:

7) #include<stdio.h> int main() { int i,j; for(i=0;i<=10;i++) { printf("\n"); for(j=0;j<=10;j++) printf(" "); printf("\n %d",j); } return 0; } Answer:	8) #include<stdio.h> int main() { int num = -3; if (num>0) printf("%d",num); else printf("%d",-num); return 0; } Answer:
9) #include<stdio.h> int main() { int x=1,y=1; for(;y;printf("%d %d\n",x,y)) y = x++ <= 5; return 0; } Answer:	10) #include<stdio.h> int main() { while(1); printf("Hi"); return 0; } Answer:

5. Write c program using for loop to print the following pattern

a) <pre> 1 12 123 1234 12345 </pre>	b) <pre> 12345 1234 123 12 1 </pre>
c) <pre> 1 22 333 4444 55555 </pre>	d) <pre> 54321 4321 321 21 1 </pre>

Tutorial 7
for loop, while loop, do while loop Contd..

1. What value of X in the below code, will print exactly 5 lines to standard output?

```
int count = 0, X;
while ( ++count < X )
{
    printf("%d\n", count);
}
```

Answer:

2. What will be the output of the below code snippet?

```
1) #include<stdio.h>
int main( )
{
    int a = 0, i = 0;
    for (i = 0; i < 5; i++)
    {
        a++;
        if (i == 3)
            break;
    }
    printf("%d", a);
    return 0;
}
```

Answer:

```
2) #include<stdio.h>
int main( )
{
    int i = 0;
    while (i < 10)
    {
        i++;
        if(i==5)
            break;
        printf("%d ", i);
    } return 0;
}
```

Answer:

```
3) #include<stdio.h>
int main( )
{
    int i = 0;
    while ( i < 10 )
    {
        i++;
        if(i==5)
            continue;
        printf("%d ", i);
    }
    return 0;
}
```

Answer:

```
4) #include<stdio.h>
int main( )
{
    int x=25,y=1;
    do
    {
        if(x>5)
            printf(" ONE");
        else if(x>10)
            printf(" TWO");
        else if(x==25)
            printf(" THREE");
        else
            printf(" FOUR");
    } while(y--);
    return 0;
}
```

Answer:

```
5) #include<stdio.h>
int main( )
{
    int i;
    for(i=0;i<=4;i++)
    {
        printf("%d",i);
        break;
        printf("Label 1");
    }
    printf("Label 2");
    return 0;
}
```

Answer:

```
6) #include<stdio.h>
int main( )
{
    int i=0,n=10;
    while(i==0)
    {
        if(n<10)
            break;
        n--;
    }
    printf("i=%d and n=%d",i,n);
    return 0;
}
```

Answer:

3. Write the output of the following code:

```
#include <stdio.h>
int main( )
{
    int x = 1 ;
    while ( x == 1 )
    {
        x = x - 1 ;
        printf ( "\n%d", x ) ;
    } return 0; }
```

Output:

4. Write the output of the following code:

```
#include <stdio.h>
int main()
{
    int x = 4, y = 0 ;
    while ( x >= 0 )
    {
        x-- ;
        y++ ;
        if ( x == y )
            continue ;
        else
            printf ( "\n%d %d", x, y ) ;
    }
    return 0;
}
```

Output:

5. Write the output of the following code:

```
#include <stdio.h>
int main()
{
    int x = 4, y = 0 ;
    while ( x >= 0 )
    {
        if ( x == y )
            break ;
        else
            printf ( "\n%d %d", x, y ) ;
        x-- ;
        y++ ;
    }
    return 0;
}
```

Output:

Tutorial 8
One dimensional arrays

1. If an array is declared as `arr[ ]={1,3,5,7,9}`; then what is the value of `arr[3]` ?

Answer:

2. _____ is the process that checks every element in the list sequentially until the desired element is found.

Answer:

3. The variable used as a subscript in an array is popularly known as _____ variable.

Answer:

4. What is the index number of the last element of an array with 29 elements?

Answer:

5. `Arr[3]=10`; initializes the _____ element of the array with value 10.

6. **State whether the following statements are true or false**

(a) According to array concept, we are allocating memory for array at compile time.

Answer:

(b) The declaration `int X[2]= {1,2,3}`; is illegal **Answer:**

(c) An Expression that evaluates to an integral value may be used as a subscript to an array variable **Answer:**

(d) In declaring an array, the array size can be a constant or variable or expression

Answer:

(e) When we use expression as a subscript, its result should be always greater than zero

Answer:

(f) An array stores all its data elements in non-consecutive memory locations

Answer:

(g) Linear Search is performed on Sorted array

Answer:

7. **What is right way to initialize array?**

(a) `int num[6] = { 2, 4, 12, 5, 45, 5 };`

(b) `int n{} = { 2, 4, 12, 5, 45, 5 };`

(c) `int n{6} = { 2, 4, 12 };`

(d) `int n(6) = { 2, 4, 12, 5, 45, 5 };`

8. What will be printed after execution of the following code?

```
int main( )
{
    int arr[10] = {1,2,3,4,5};
    printf("%d", arr[5]);
    return 0; }
```

- a. Garbage Value
- b. 5
- c. 6
- d. 0
- e. None of these

9. Write the Output for the following code:

```
#include<stdio.h>
int main( )
{
    int arr[5], i=-1, z;
    while(i<5)
        arr[i]=++i;
    for(i=0; i<5; i++)
        printf("%d, ", arr[i]);
    return 0;
}
```

10. Find the output for the below given code snippets.

```
a) #include<stdio.h>
int main( )
{
    int m[ ] = {1,2,3,4,5};
    int x, y=0;
    for( x=0;x<5;x++)
    {
        y=y+m[x];
    }
    printf("%d", y);
    return 0;
}
```

Answer:

```
b) #include<stdio.h>
int main( )
{
    int arr[10] = {1,2,3,4,5};
    printf("%d", arr[5]);
    return 0;
}
```

Answer:

```
c) #include<stdio.h>
int main( )
{
    int a[ ]={0,1,2,0,1,2,0,1,2};
    printf("\n%d",a[3]);
    printf("\n%d",a[a[3]]);
    printf("\n%d",a[a[3]+a[1]]);
    return 0;
}
```

Answer:

```
d) #include<stdio.h>
int main( )
{
    int a[5] = {5, 1, 15, 20, 25};
    int i, j, m;
    i = ++a[1];
    j = a[1]++;
    m = a[i++];
    printf("%d, %d, %d", i, j, m);
    return 0;
}
```

Answer:

Tutorial 9 **Strings**

1. Assuming the variable string contains the value “The sky is the limit.”, determine what output of the following program segments will be.

Instruction	Output
a) printf(“%s”,string);	The sky is the limit.
b) printf(“%25.10s”,string);	
c) printf(“%s”,&string[0]);	
d) for(i=0;string[i]!='\0';i++) printf(“%c”,string[i]);	
e) for(i=0;string[i]!='\0';i++) printf(“%d\n”,string[i]);	

2. Assume that s1,s2 and s3 are declared as follows:

char s1[10]=”he”, s2[20]=”she”, s3[30], s4[30];

What will be the output of the following statements executed in sequence?

<pre>printf(“%s\n”,strcpy(s3,s1)); printf(“%s\n”,strcat(strcat(strcpy(s4,s1),”or”),s2)); printf(“%d\n%d\n”,strlen(s2)+strlen(s3),strlen(s4));</pre>	Output:
---	----------------

3. Which of the following function is used to find the first occurrence of a given string in another string?

- | | |
|-------------|--------------|
| A. strchr() | B. strrchr() |
| C. strstr() | D. strlen() |

4. Find the output for the following:

<pre>a. #include<stdio.h> int main() { char str1[]={'H','I'}; char str2[]={'H','I','\0'}; if(strcmp(str1,str2)==0) printf("\n The strings are equal"); else printf("\n Strings are not equal"); return 0; }</pre> Answer:	<pre>b. #include<stdio.h> int main() { char str[]="Programming in C"; int i; while(str[i]!='\0') { if(i%2==0) printf("%c",str[i]); i++; } return 0; }</pre> Answer:
<pre>e. #include<stdio.h> int main() { char str1[]="Programming"; char str2[]="Project"; printf("%d",strncmp(str1,str2,3)); return 0; }</pre> Answer:	<pre>d. #include<stdio.h> int main() { char ch='k'; printf("%c",ch+10); return 0; }</pre> Answer:

5. If the two strings are identical, then strcmp() function returns

- 1
- 1
- 0
- yes

6. Find errors in the following codes.

<pre>a. #include<stdio.h> int main() { char str1[]="Programming"; char str2[]="In C"; str1=str2; printf("\n str1=%s",str1); return 0; }</pre> Answer:	<pre>b. #include<stdio.h> int main() { char str[]={'H','E','L','L','\0'}; printf("\n str=%s",str); return 0; }</pre> Answer:
--	--

```
c. #include<stdio.h>
int main( )
{
    char str[10];
    strcpy(str,"Hello there");
    printf("\n str=%s",str);
    return 0;
}
```

Answer:

```
d. #include<stdio.h>
int main( )
{
    char str[5]="HELLO";
    printf("\n str=%s",str);
    return 0;
}
```

Answer:

7. Fill in the blanks:

- Strings are _____
- Every string is terminated with a _____
- _____function terminates reading input as soon as it finds a blank space in the string.
- The function atoi () is present in_____header file.
- If the string is given as "ABCD", the length of this string is ____.

8. State True or False

- A string Hello World can be read using scanf().
- The function strcmp("and","ant") will return a positive value.
- The gets() and scanf() automatically appends a null character at the end of the string read from the keyboard.

Tutorial 10
User defined functions

1. Which of the following prototype declarations are invalid?

a. int (fun) void;	
b. double fun(void)	
c. float fun(x,y,n);	
d. void fun(void, void);	
e. int fun(int a, b);	
f. fun(int,float, char);	

2. Find errors in the following function calls:

a. void xyz();	
b. xyx(void);	
c. xyx(int x, int y);	
d. xyzz();	
e. xyz()+xyz();	

3. A function to divide two floating point numbers is as follows:

```
float divide (float x, float y)
{
    return(x/y);
}
```

What will be the value of the following function calls?

- a. divide(10,2)
- b. divide(9,2)
- c. divide(4.5,1.5)
- d. divide(2.0,3.0)

4. Which of the following header lines are invalid? Why?

a. float average(float x, float y, float y)	
b. double power(double a, int n-1)	
c. int product(int m, 10)	
d. double minimum(double x; double y)	
e. int mul(int x, y)	

5. The keyword used to transfer control from main function back to the calling function is

- a. switch
- b. goto
- c. return
- d. break

6. A function cannot be defined inside another function

- a. True
- b. false

7. Use of functions

- a. Helps to avoid repeating a set of statements many times.
- b. Enhances the logical clarity of the program.
- c. Makes the debugging task easier.
- d. All of the above

8. Write the output for the following code:

```
int main( )
{
    int i = abc(10);
    printf("%d", --i);
    return 0;
}
```

```
int abc(int i)
{
    return(i++);
}
```

- a. 10
- b. 9
- c. 11
- d. None of these.

9. What is function?

- a. Function is a block of statements that perform some specific task.
- b. Function is the fundamental modular unit. A function is usually designed to perform a specific task.
- c. Function is a block of code that performs a specific task. It has a name and it is reusable.
- d. All of the above

10. Pick the correct statements.

- I. The body of a function should have only one return statement.
 - II. The body of a function may have many return statements.
 - III. A function can return only one value to the calling environment.
 - IV. If return statement is omitted, then the function does its job but returns no value to the calling environment.
- a. I and II
 - b. I and III
 - c. II and III
 - d. II and IV
 - e. III and IV

11. What will be printed when this program is executed?

```
int f(int x)
{
    if(x <= 4)
        return x;
    return f(--x);
}
int main()
{
    printf("%d ", f(7));
    return 0;
}
```

- a. 4 5 6 7
- b. 1 2 3 4
- c. 4

- d. Syntax error
- e. Runtime error

12. The function scanf() returns -----

- a. The actual values read for each argument.
- b. 1
- c. 0
- d. The number of successful read input values.
- e. ASCII value of the input read.

13. Find errors, if any in the following definitions:

a. void abc(int a, int b) <pre>{ int c; return c; }</pre> Answer:	b. int abc(int a, int b) <pre>{ }</pre> Answer:
c. int abc(int a,int b) <pre>{ double c=a+b; return (c); }</pre> Answer:	d. void abc(void) <pre>{ return; }</pre> Answer:

TUTORIAL 11
Structures, Pointers

- 1. Define a structure named abc with three members of type int, float and char respectively. Use this definition to identify valid declaration and initialization statements given below. Justify your answers.**

- a. struct a,b,c
- b. struct abc a,b,c
- c. abc x,y,z;
- d. struct abc a[];
- e. struct abc a={ };
- f. struct abc=b,(1+2,3.0,"xyz")
- g. struct abc c={4,5.0,'G'};

- 2. Given the declaration**

struct abc a,b,c;

Which of the following statements are legal and why?

- a. scanf("%d",&a);
- b. printf("%d", b);
- c. a=b;
- d. a=b+c;

- 3. Which of the following are correct statements for declaring one dimensional array of structures of type struct item_bank?**

- a. int item_bank items[10];
- b. struct items[10] item_bank;
- c. struct item_bank items[10];
- d. struct items item_bank[10];
- e. struct item_bank items(10);

- 4. If m and n have been declared as integers and p1 and p2 as pointers to integers, then state errors, if any in the following statements.**

- a. p1=&m;
- b. p2=n;
- c. *p1=&n;
- d. p2=&*&m;
- e. m=p2-p1;

5. Write the output for the following code.

```
#include <stdio.h>
int main( )
{
    int *ptr;
    int x;
    ptr = &x;
    *ptr = 0;
    printf(" x = %dn", x);
    printf(" *ptr = %dn", *ptr);
    *ptr += 5;
    printf(" x = %dn", x);
    printf(" *ptr = %dn", *ptr);
    (*ptr)++;
    printf(" x = %dn", x);
    printf(" *ptr = %dn", *ptr);
    return 0;
}
```

6. Write the output for the following code.

```
#include <stdio.h>
int main( )
{
    float arr[5] = {12.5, 10.0, 13.5, 90.5, 0.5};
    float *ptr1 = &arr[0];
    float *ptr2 = ptr1 + 3;
    printf("%f ", *ptr2);
    printf("%d", ptr2 - ptr1);
    return 0;
}
```

7. Write the output for the following code.

```
#include<stdio.h>
int main()
{
    int *ptr, a=10;
    ptr = &a;
    *ptr += 1;
    printf("%d, %d", *ptr, a);
    return 0;
}
```

8. * is called as _____.

- a. Address Operator
- b. Value at Operator
- c. Scope Resolution Operator
- d. None of these

9. Write the output for the following

```
#include<stdio.h>
int main( )
{
    int i=3, *j, k;
    j = &i;
    printf("%d\n", i * *j * i ++*j);
    return 0;
}
```

- a. 30
- b. 27
- c. 9
- d. 3

10. Write the output for the following

```
#include<stdio.h>
int main( )
{
    int x=30, *y, *z;
    y=&x; /* Assume address of x is 500 and integer is 4 byte size */
    z=y;
    *y++=*z++;
    x++;
    printf("x=%d, y=%d, z=%d\n", x, y, z);
    return 0;
}
```

- a. x=31, y=502, z=502
- b. x=31, y=500, z=500
- c. x=31, y=498, z=498
- d. x=31, y=504, z=504

11. User-defined data type can be derived by_____.

- a. Struct
- b. Typedef
- c. Enum
- d. All the mentioned above

12. Which of the following operators can be applied on structure variables?

- a. Equality comparison (==)
- b. Assignment (=)
- c. Both of the above
- d. None of the above

13. The following program reports an error on compilation.

```
#include<stdio.h>
int main()
{
    float i=10, *j;
    void *k;
    k=&i;
    j=k;
    printf("%f", *j);
    return 0;
}
```

- a. True
- b. False

14. The operator used to get value at address stored in a pointer variable is

- a. *
- b. &
- c. &&
- d. ||

TUTORIAL 12
FILES**Answer the Following Questions :-**

1. Differentiate between gets() and fgets().
2. List different modes in which a file can be opened in a C program.
3. _____ is the stream where a program writes its output data.
4. A file must be opened in _____ mode if it is being opened for updating.
5. _____ header files is used to perform File I/O operations.
6. By default, all the files are opened in _____ mode.
7. Write the syntax and description of the following file functions.

File Operation	Declaration& Description
fopen()-to open a file	
fclose()-to close a file	
fgets()-to read a file	
fprintf()-to write into file	
fseek()-sets position to a desired point in a file	

8. What operation can be performed on the file "NOTES.TXT" using the below code?

```
FILE *fp;
```

```
fp = fopen("NOTES.TXT", "r+");
```

9. Which of the given options tells about the use of eof() stream function ?
- a. Returns true if a file open for reading has reached the next character.
 - b. Returns true if a file open for reading has reached the next word.
 - c. Returns true if a file open for reading has reached the end.
 - d. Returns true if a file open for reading has reached the middle.